

G-CAF

GovDigital Composable Architecture Framework

Volume 5

Reference Architectures

v2.0

Table of Contents

Chapter 1 Reference Architectures Overview

Purpose, catalog, and template system

Chapter 2 REF-001: OneHub — Overview

G2E unified employee experience platform

Chapter 3 REF-001: OneHub — Architecture & PBCs

10 composition platform PBCs and composition contracts

Chapter 4 REF-001: OneHub — Channels & Operations

Omni-channel, ADRs, observability, and BCM

Chapter 5 REF-002: GovData Platform — Overview

Government Data Fabric and Analytics Platform

Chapter 6 REF-002: GovData Platform — Architecture & PBCs

15 platform PBCs across 5 layers

Chapter 7 REF-002: GovData Platform — Patterns & Operations

Data mesh, medallion, observability, BCM

Chapter 8 REF-003: IDP — Internal Developer Platform

AI-enabled developer platform with MCP servers

Chapter 1

Reference Architectures Overview

Reference architectures demonstrate G-CAF principles in concrete, platform-specific implementations. They are proven blueprints that entities can adopt directly or adapt.

Reference Architecture Catalog

ID	Name	Description	Files	Version
REF-001	OneHub	Unified G2E omni-channel employee experience platform	47	v1.13
REF-002	GovData Platform	Enterprise data fabric, analytics, and AI platform	56	v1.17
REF-003	IDP	AI-enabled Internal Developer Platform with MCP servers	12	v1.0.0

Each reference architecture includes an overview, detailed architecture, platform PBCs, ADRs, channels/integrations, implementation guide, operations specifications, and more.

Template System

New reference architectures are created using a standardized template system with core templates (index, overview, architecture, PBCs, implementation), extended templates (operations frame aligned), guidance documents, and reusable theory modules.

Creating a New Reference Architecture

- Copy the template structure from the templates folder
- Assign the next available ID (G-CAF-REF-NNN)
- Complete core templates first, then extended templates as needed
- Submit for DDA review via the gateway process

Chapter 2

REF-001: OneHub — Overview

OneHub is the **Government-to-Employee (G2E) Composition Platform** for Abu Dhabi Government, providing a unified omni-channel experience for government employees.

The single digital workplace for Abu Dhabi Government employees — anticipating needs, automating routine tasks, and empowering focus on high-value work.

Scope

In Scope	Description
Employee Self-Service	HR, finance, and administrative transactions
Unified Inbox	Tasks, approvals, and notifications from all PBCs
Federated Search	Cross-PBC search for information and actions
Omni-Channel Access	Web, mobile, kiosk, chatbot, API
Personalization	Adaptive experience based on role and behavior

Service Model

Attribute	Value
Service Model	G2E (Government-to-Employee)
Target Users	Abu Dhabi Government employees
User Base	~50,000 (Year 1) → ~100,000 (Year 5)
Entities Served	All Abu Dhabi Government entities

Key Differentiators

Differentiator	Description
Composition-First	Assembles PBCs, doesn't rebuild capabilities
Omni-Channel	Same experience across web, mobile, kiosk, chatbot, API
AI-Ready	Architecture supports AI agent integration
NS-Compliant	Follows Normalized Systems principles
Employee-Centric	Designed around employee journeys, not systems

Success Metrics

Metric	Target
Employee Adoption	95% active users
Self-Service Rate	90% transactions via self-service
Channel Availability	99.9% uptime
Task Completion Rate	85% within SLA
Employee NPS	+50

Chapter 3

REF-001: OneHub — Architecture & PBCs

OneHub's architecture follows the composition pattern: channels connect through an Omni-Channel Gateway to Composition Platform PBCs, which consume business PBCs via composition contracts through GovDX.

G-CAF Layer Positioning

G-CAF Layer	OneHub Role
Experience Tier	OneHub IS the composition platform
Capability Tier	OneHub CONSUMES PBCs (HR, Finance, Productivity)
Platform Tier (GovDX)	Integration via API Gateway, Event Bus, Schema Registry
Platform Tier (DPI)	Consumes Digital Signatures, Payments, Registries
Platform Tier (Cloud)	Runs on Abu Dhabi Government Cloud
Operations Frame	Implements all operations standards

10 Composition Platform PBCs

PBC ID	Name	Type	Purpose
OH-CPL-001	Experience Shell	Application	Portal container, layout, theming
OH-CPL-002	Omni-Channel Gateway	Orchestration	Channel adaptation and routing
OH-CPL-003	Widget Registry	Data	Catalog of available UI components
OH-CPL-004	Universal Inbox	Orchestration	Aggregated tasks and notifications
OH-CPL-005	Federated Search	Data	Cross-PBC search federation
OH-CPL-006	Navigation Orchestrator	Orchestration	Routing, deep linking, breadcrumbs
OH-CPL-007	Context Manager	Application	Session context, user preferences
OH-CPL-008	Personalization Engine	Analytics	Content ranking, recommendations
OH-CPL-009	Composition Config	Data	PBC composition configuration
OH-CPL-010	Channel Presence	Application	Online status, device management

Composition Contracts

Business PBCs (HR, Finance, Productivity) contribute to OneHub by implementing composition contracts: UI Contribution, Task Contribution, Search Contribution, Navigation Contribution, Notification Contribution, and

Context Contribution.

Chapter 4

REF-001: OneHub — Channels & Operations

Channels

Channel	Technology	Use Case
Web Portal	React micro-frontends in Backstage	Primary desktop access for all employees
Mobile App	React Native / PWA	On-the-go approvals, notifications, quick tasks
Kiosk Terminal	Simplified web with touch UI	Shared workstations, service centers
Chatbot Agent	AI conversational agent (Agent PBC)	Natural language task execution, FAQ
API Direct	REST/GraphQL via GovDX	System integrations, automation
WhatsApp Bot	WhatsApp Business API	Notifications, simple approvals

Architecture Decision Records (ADRs)

ADR	Decision	Rationale
ADR-001	Micro-frontend composition	Independent PBC UI deployment; team autonomy
ADR-002	Azure-native implementation	AD Gov Cloud alignment; managed services
ADR-003	Entra ID for identity	Existing government SSO; MFA support
ADR-004	Federated search pattern	Cross-PBC search without centralized index
ADR-005	Event-driven notifications	Real-time updates; loose coupling

Operations Specifications

OneHub includes detailed specifications for observability (metrics, logs, traces, dashboards), security operations (SOC integration, SIEM, incident response), secrets management (Key Vault, rotation), capacity sizing (compute, storage, scaling), business continuity (DR, backup, RTO/RPO), and FinOps (cost allocation, metering).

Additional Specifications

Document	Purpose
Hub-Spoke Topology	Network and subscription layout
Adoption Tiers	Phased rollout: Foundation → Enhanced → Advanced
Skills Matrix	Required team capabilities

Document	Purpose
Localization	Arabic/English RTL support
Technical Debt	Debt tracking and remediation
Vendor Evaluation	Technology selection criteria
Sovereign Requirements	Data residency and compliance

Chapter 5

REF-002: GovData Platform — Overview

GovData Platform is the **Government Data Fabric and Analytics Platform** — providing unified data access, governance, quality management, and analytics across all entities and PBCs.

The single source of trusted government data — enabling data-driven decisions, AI-powered services, and seamless data sharing while ensuring privacy, compliance, and sovereignty.

Scope

In Scope	Description
Data Catalog and Discovery	Enterprise-wide data asset discovery and documentation
Active Metadata Management	Automated metadata collection, classification, enrichment
Data Quality Management	Quality rules, profiling, monitoring, remediation
Data Lineage	End-to-end data flow tracking and change impact
Reference Data Management	Master data, lookups, code lists
Data Virtualization	Unified access without data movement
Knowledge Graphs	Entity relationships and semantic modeling
Analytics Data Store	Prepared data for ML, AI, and analytics
ML Feature Store	Reusable ML features for AI services
Data Access and Consent	Governed access with consent enforcement

Service Model

Attribute	Value
Service Model	G2G + G2E (internal platform)
Target Users	Data stewards, analysts, data scientists, PBC developers
Data Sources	~200 Application PBCs, 50+ legacy systems
Entities Served	All Abu Dhabi Government entities

Key Differentiators

Differentiator	Description
Data Mesh Ready	Federated ownership with centralized governance

Differentiator	Description
Active Metadata	AI-driven metadata automation and discovery
Consent-Aware	Privacy-by-design with consent enforcement
NS-Compliant	Single change driver per component
AI-First	ML feature store and knowledge graphs for AI
Zero-Copy	Data virtualization minimizes data movement

Success Metrics

Metric	Target
Data Asset Coverage	100% PBCs cataloged
Metadata Automation	80% auto-classified
Data Quality Score	95% pass rate
Lineage Coverage	100% critical entities

Chapter 6

REF-002: GovData Platform — Architecture & PBCs

GovData's architecture is organized in 5 layers: Data Access, Governance, Semantic, Data Quality, and Data Serving — all consuming from data sources via GovDX Event Bus.

Architecture Layers

Layer	Purpose	Key PBCs
Data Access	Secure, governed entry point for all consumer data access	Data Access Gateway
Governance	Policy enforcement, ownership, classification	Data Governance Hub, Consent Manager
Semantic	Discovery, understanding, relationships	Data Catalog, Metadata Manager, Knowledge Graph, Data Lineage
Data Quality	Profiling, rules, monitoring, scoring	Data Quality Engine
Data Serving	Reference data, analytics, ML features	Reference Data Hub, Data Virtualization, Analytics Store, ML Feature Store

15 Platform PBCs

PBC	Type	Purpose
Data Catalog	Data	Metadata discovery and search
Data Governance Hub	Application	Policy, ownership, classification, access control
Data Quality Engine	Application	Quality rules, profiling, monitoring, remediation, scoring
Data Lineage	Data	End-to-end data flow tracking
Metadata Manager	Data	Technical metadata repository
Data Access Gateway	Orchestration	Secure data access with OAuth, RBAC, consent
Data Virtualization	Data	Federated query layer (zero-copy)
ML Feature Store	Data	ML feature management and serving
Analytics Data Store	Data	Analytical data persistence (medallion)
Reference Data Hub	Data	Master and reference data management
Consent Manager	Application	Privacy consent tracking and enforcement
Knowledge Graph	Data	Semantic relationships and ontology
Data Engineering Workbench	Application	ETL/ELT development environment

PBC	Type	Purpose
Synthetic Data Generator	Application	Privacy-safe test data generation
Runtime Security Engine	Application	Dynamic data protection and masking

G-CAF Layer Mapping

G-CAF Module	GovData Implementation
Data Asset Catalog (G-CAF-044)	Data Catalog PBC + Metadata Manager PBC
Data Contract (G-CAF-071)	Data Governance Hub PBC
Data Ownership (G-CAF-071b)	Data Governance Hub + Consent Manager PBC
Data Fabric (G-CAF-106)	All platform PBCs collectively
PBC Types (G-CAF-052)	Serves Analytics PBCs and Data PBCs

Chapter 7

REF-002: GovData Platform — Patterns & Operations

Architecture Decision Records

ADR	Decision	Rationale
ADR-001	Lakehouse architecture	Combines data lake flexibility with data warehouse structure
ADR-002	Data mesh governance	Federated domain ownership; centralized platform
ADR-003	Microsoft Purview	AD Gov Cloud alignment; unified governance
ADR-004	Unity Catalog	Cross-engine metadata; fine-grained access
ADR-005	Medallion pattern	Bronze/Silver/Gold data quality progression

Data Patterns

Data Mesh: Federated domain ownership where each domain owns its data products, with centralized governance via the Data Governance Hub. Domains publish data products through standardized contracts.

Medallion Architecture: Three-layer data quality progression — Bronze (raw ingestion), Silver (cleansed, validated), Gold (business-ready, aggregated). Each layer has clear quality SLAs.

Data Virtualization: Federated query layer enabling zero-copy data access. Consumers query data through virtual views without ETL pipelines, reducing data movement and staleness.

Operations Specifications

The GovData Platform includes comprehensive specifications for hub-spoke network topology, observability (data pipeline monitoring, quality dashboards), security operations, secrets management, capacity sizing (storage tiers, compute scaling), business continuity, FinOps, adoption tiers, skills matrix, localization, technical debt management, and vendor evaluation criteria.

Additional Documents

Document	Purpose
Data Product Lifecycle	End-to-end data product management
BI Semantic Layer	Business-friendly data abstraction for reporting
Data Exchange Platform	Cross-entity data sharing governance
Domain Model Alignment	Mapping data domains to G-CAF bounded contexts

Document	Purpose
Data Modeling Tools	Approved tools and standards for data modeling
Composition Contribution	How GovData PBCs contribute to OneHub
Environment Specs	DEV/SIT/UAT/STAGING/PROD specifications
Gap Analysis	Current state vs. target architecture assessment

Chapter 8

REF-003: Internal Developer Platform (IDP)

The IDP reference architecture defines the implementation blueprint for an **AI-enabled Internal Developer Platform** aligned with G-CAF governance.

Executive Summary

Aspect	Value
Purpose	Accelerate development through self-service and AI assistance
Scope	All G-CAF governed entities
Key Benefit	PBC creation in 2 days vs. 2 weeks
AI Capability	Natural language platform interactions
Governance	Full G-CAF alignment with automated enforcement

Developer Interfaces

Interface	Description
Web UI (Backstage)	Service catalog, documentation, scorecards
CLI	Command-line PBC scaffolding and management
IDE Extension	In-editor golden path guidance and validation
Chat ("Ask IDP")	Natural language platform interaction (AI Agent PBC)
Slack/Teams	ChatOps for notifications, approvals, quick actions

Platform PBCs

PBC ID	Name	Purpose
PBC-IDP-001	Software Catalog	Entity registry (PBCs, APIs, teams, infrastructure)
PBC-IDP-002	Golden Path Engine	Template scaffolding with G-CAF compliance built in
PBC-IDP-003	Scorecard Service	Automated PBC health and compliance scoring
PBC-IDP-004	Environment Provisioner	Self-service infrastructure provisioning
PBC-IDP-005	TechDocs Service	Documentation-as-code with auto-generation
PBC-IDP-006	IDP Assistant	AI Agent PBC for natural language platform interactions

MCP Server Adapters

The IDP uses Model Context Protocol (MCP) servers to enable AI integration across platform capabilities: catalog-mcp (entity queries), golden-path-mcp (template scaffolding), infra-mcp (infrastructure provisioning), pipeline-mcp (CI/CD management), and docs-mcp (documentation generation).

AI Integration Patterns

Pattern	Description
AIOps Integration	AI-driven monitoring, alerting, and auto-remediation
Context Engineering	Structured context provision for LLM interactions
Guardrails	Safety boundaries for AI agent actions on infrastructure
Natural Language Workflows	Self-service platform tasks via conversational AI

IDP Foundation

Component	Technology
Portal Shell	Backstage (open source developer portal)
Database	PostgreSQL Cluster
Vector Database	For AI semantic search and context
MCP Gateway	Model Context Protocol routing
LLM Gateway	AI model management and safety